

A Level Mathematics 9709 · P3 Pure Mathematics 3

全套中英文详细学习手册

Detailed bilingual student manual · concept, derivation, worked examples
and exam method

本册共 10 个独立课时。每课按四页展开：前两页依次讲解 8 个编号知识点；后两页完成 9 项编号化逐题精讲、答案、纠错与评价。题目均为依据考纲原创的训练示例，不复制历年试题。

使用范围与版本 | Scope

0580 册依据 Cambridge IGCSE Mathematics 0580 (2025/2027) 的 Core/Extended 分层；9709 册依据 Cambridge International AS & A Level Mathematics 9709 (2026/2027)。M2 单独标为拓展/旧体系衔接材料，不作为现行独立考试组件。

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第 1 课 | 多项式与部分分式 | Polynomials and partial fractions

Lesson 1 of 10 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

因式定理、余式定理与多项式除法用于拆解高次式；部分分式把有理式化为可积分形式。
学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. The factor theorem, polynomial division and partial fractions form one chain: factor the denominator, determine unknown numerators, verify the identity, and only then use the decomposition.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

余式定理 $p(a)$ 等于多项式除以 $x-a$ 的余数；若 $p(a)=0$ ，则 $x-a$ 是因式。部分分式前必须先确认分子次数低于分母。
公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

先确认真分式；分母不同因子决定分子形式，重复因子需逐级列项。
不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

把分母完全因式分解：不同一次因子用常数分子；重复因子逐次列
 $A/(x-a)+B/(x-a)^2$ ；不可约二次因子使用线性分子。
若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 1 课（知识点续） | 多项式与部分分式 | Polynomials and partial fractions

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

代回原式或通分复核系数。积分前若区间含分母零点，必须检查积分是否存在，不能机械套部分分式。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

分解 $(5x+1)/[(x+1)(x+2)]$ 。正确答案:

令 $A/(x+1)+B/(x+2)$ ，得 $5x+1=A(x+2)+B(x+1)$ 。代 $x=2$ 得 $B=11/3$ ； $x=-1$ 得 $A=4/3$ 。

完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

可使用以下骨架组织答案: From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

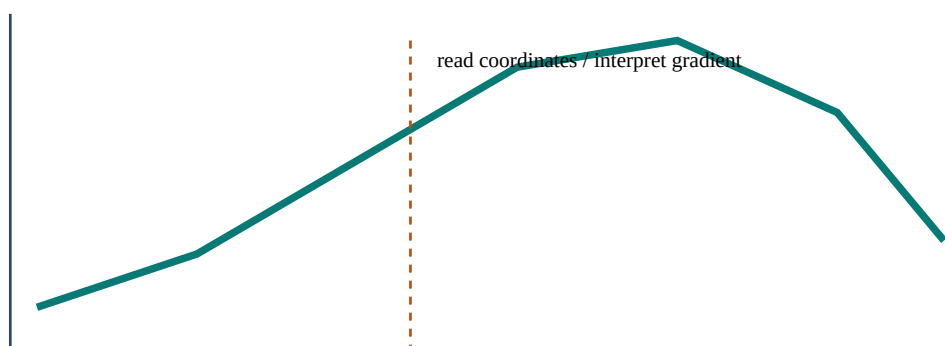
English explanation. The factor theorem, polynomial division and partial fractions form one chain: factor the denominator, determine unknown numerators, verify the identity, and only then use the decomposition. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Find remainder when $p(x)=x^2x+5$ is divided by x^2 . 参考答案: Remainder= $p(2)=84+5=9$.

不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: Remainder= $p(2)=84+5=9$.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 1 课（逐题精讲） | 多项式与部分分式 | Polynomials and partial fractions

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. 分解 $(5x+1)/[(x+1)(x+2)]$ 。

正确答案 | Correct answer.

令 $A/(x+1)+B/(x+2)$ ，得 $5x+1=A(x+2)+B(x+1)$ 。代 $x=2$ 得 $B=11/3$ ； $x=-1$ 得 $A=4/3$ 。评分展开：

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Find remainder when $p(x)=x^2x+5$ is divided by $x+2$.

参考答案 | Model answer. Remainder= $p(-2)=8+5=13$. 得分点：

正确识别对象与条件；选择有效方法；展示关键运算；用完整句回答题目。

第 1 课（精讲续） | 多项式与部分分式 | Polynomials and partial fractions

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 2 课 | 模、指数与对数深化 | Modulus, exponential and logarithmic functions

Lesson 2 of 10 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

模函数图像可由反射负值部分得到；指数增长率与当前值成正比。

学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. Modulus removes sign, logarithms require positive arguments, and exponential outputs are positive. These facts determine both the method and the admissible solution set.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

$|f(x)|$ 表示 $f(x)$ 到 0 的距离；指数函数始终为正；对数的真数必须大于 0。这三个定义域条件是方程验根的核心。公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

解方程要结合图像交点和定义域；换底处理不同底数。

不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

模方程按 $f(x)=a$ 与 $f(x)=-a$ 分支；指数方程统一底数或取对数；对数方程先合并，再检查每个真数。

若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 2 课（知识点续） | 模、指数与对数深化 | Modulus, exponential and logarithmic functions

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

平方和取对数都可能改变原方程允许范围。最后必须把候选根代回原式，而不是只检查变形后的方程。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

解 $e^{(2x)}=7$ 。正确答案: $2x=\ln 7$, $x=0.5\ln 7$ 。

完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

可使用以下骨架组织答案: From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

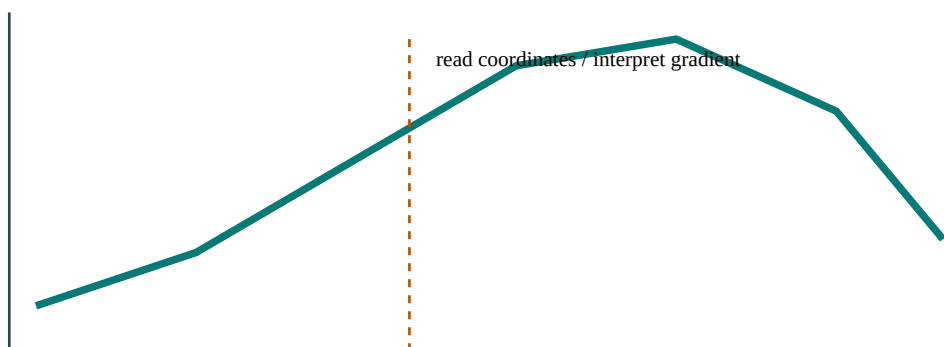
English explanation. Modulus removes sign, logarithms require positive arguments, and exponential outputs are positive. These facts determine both the method and the admissible solution set. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Solve $|x^4|=3$. 参考答案: $x^4=3$ gives $x=\pm\sqrt[4]{3}$; $x^4=3$ gives $x=\pm 1$.

不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: $x^4=3$ gives $x=\pm\sqrt[4]{3}$; $x^4=3$ gives $x=\pm 1$.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 2 课（逐题精讲） | 模、指数与对数深化 | Modulus, exponential and logarithmic functions

逐题精讲与评价拓展 | Exam workshop (共 9 项)

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. 解 $e^{(2x)}=7$ 。

正确答案 | Correct answer. $2x=\ln 7$, $x=0.5\ln 7$ 。评分展开:

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Solve $|x+4|=3$.

参考答案 | Model answer. $x+4=3$ gives $x=\pm \sqrt{7}$; $x+4=3$ gives $x=\pm 1$. 得分点:

正确识别对象与条件; 选择有效方法; 展示关键运算; 用完整句回答题目。

第 2 课（精讲续） | 模、指数与对数深化 | Modulus, exponential and logarithmic functions

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 3 课 | 三角函数深化 | Advanced trigonometry

Lesson 3 of 10 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

辅助角把 $a\sin x + b\cos x$ 化成 $R\sin(x + \alpha)$; 恒等变换为方程和最大最小值服务。

学习时先用自己的话重述定义, 再指出它与相邻概念的区别; 这样遇到改变表述的题目仍能识别本质。

English explanation. The auxiliary-angle form reveals amplitude, phase shift, range and solvability. A principal calculator value is only the beginning of solving a trigonometric equation.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

辅助角公式把 $a\sin x + b\cos x$ 写成 $R\sin(x + \alpha)$ 或 $R\cos(x - \beta)$, 其中

$R = \sqrt{a^2 + b^2}$ 。比较系数时要保持象限和符号。

公式前先写变量含义和允许范围, 公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

$R = \sqrt{a^2 + b^2}$, 比较系数求 α , 随后处理值域。

不要把计算器按键当作证明: 至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

得到单一三角函数后, 先利用 $-1 \leq \sin \leq 1$ 确定值域或解存在条件, 再在指定区间寻找所有解。

若图形趋势、代数符号和数表不一致, 应暂停并检查模型, 而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 3 课（知识点续） | 三角函数深化 | Advanced trigonometry

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

反三角函数只给主值；完整答案必须用周期和象限补齐。题目使用弧度时不能以 degree 模式计算。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

写 $3\sin x + 4\cos x$ 为 $R\sin(x + \alpha)$ 。正确答案： $R=5$, $5\cos \alpha = 3$, $5\sin \alpha = 4$, 所以 $\alpha = \tan^{-1}(4/3)$ 。完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

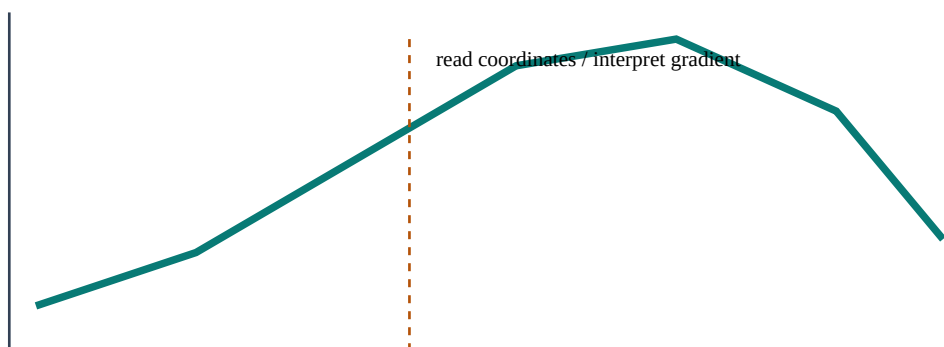
可使用以下骨架组织答案： From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

English explanation. The auxiliary-angle form reveals amplitude, phase shift, range and solvability. A principal calculator value is only the beginning of solving a trigonometric equation. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Hence state the maximum value. 参考答案： Since $\sin(x + \alpha) \leq 1$, the maximum is 5. 不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: Since $\sin(x + \alpha) \leq 1$, the maximum is 5.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 3 课（逐题精讲） | 三角函数深化 | Advanced trigonometry

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. 写 $3\sin x + 4\cos x$ 为 $R\sin(x + \alpha)$ 。

正确答案 | Correct answer. $R=5$, $5\cos \alpha=3$, $5\sin \alpha=4$, 所以 $\alpha=\tan^{-1}(4/3)$ 。评分展开：
先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Hence state the maximum value.

参考答案 | Model answer. Since $\sin(x + \alpha) \leq 1$, the maximum is 5. 得分点：
正确识别对象与条件；选择有效方法；展示关键运算；用完整句回答题目。

第 3 课（精讲续） | 三角函数深化 | Advanced trigonometry

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 4 课 | 复杂微分 | Further differentiation

Lesson 4 of 10 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

隐函数、参数方程和二阶导数扩展变化率分析。

学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. Implicit and parametric differentiation preserve the dependency between variables. Every differentiated y -term carries dy/dx , and parametric gradients are ratios of rates.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

隐函数中 y 是 x 的函数，因此微分 y 得 $2y(dy/dx)$ 。参数方程则分别求 dx/dt 与 dy/dt ，再相除。

公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

隐式微分每个 y 项乘 dy/dx ；参数式 $dy/dx = (dy/dt)/(dx/dt)$ 。

不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

二阶导可通过 $d/dx(dy/dx)$ 获得；参数式二阶导要把 $d/dt(dy/dx)$ 再除以 dx/dt 。

若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 4 课（知识点续） | 复杂微分 | Further differentiation

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

$dx/dt=0$ 时 dy/dx 的表达式可能无定义，对应竖直切线或需进一步检查，不能直接约去使其消失。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

$x+y=25$ ，求 dy/dx 。正确答案： $2x+2y \ dy/dx=0$ ，所以 $dy/dx=x/y$ 。
完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

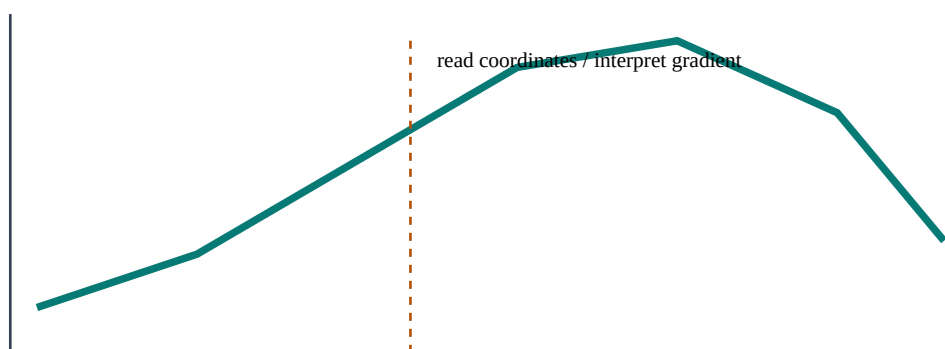
可使用以下骨架组织答案：From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

English explanation. Implicit and parametric differentiation preserve the dependency between variables. Every differentiated y -term carries dy/dx , and parametric gradients are ratios of rates. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Given $x=t+1, y=t$, find dy/dx 。参考答案： $dy/dx=(3t)/(2t)=3t/2$, for $t \neq 0$ 。
不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: $dy/dx=(3t)/(2t)=3t/2$, for $t \neq 0$.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 4 课（逐题精讲） | 复杂微分 | Further differentiation

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. $x+y=25$ ，求 dy/dx 。

正确答案 | Correct answer. $2x+2y \, dy/dx=0$ ，所以 $dy/dx=x/y$ 。评分展开：

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Given $x=t+1, y=t$, find dy/dx .

参考答案 | Model answer. $dy/dx=(3t)/(2t)=3t/2$, for $t \neq 0$. 得分点：

正确识别对象与条件；选择有效方法；展示关键运算；用完整句回答题目。

第 4 课（精讲续） | 复杂微分 | Further differentiation

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 5 课 | 复杂积分 | Further integration

Lesson 5 of 10 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

换元、分部积分和部分分式处理非直接积分。

学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. Substitution reverses the chain rule, integration by parts reverses the product rule, and partial fractions turn one rational expression into simpler integrable terms.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

换元法寻找“内函数及其导数”；分部积分来自乘积法则；部分分式处理有理函数。方法选择比计算本身更重要。公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

看到复合及其导数考虑换元；多项式 $\times$ 指数/三角常用分部积分。

不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

定积分换元时可以同时换上下限；若保留原上下限，则积分结束后必须换回原变量。

若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 5 课（知识点续） | 复杂积分 | Further integration

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

面积问题要先确定上下曲线和交点。有符号积分为负不等于几何面积为负，必要时分区间取正。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

计算 $\int x e^x dx$ 。正确答案：取 $u=x, dv=e^x dx$ ，得 $x e^x - \int e^x dx = x e^x - e^x + C$ 。
完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

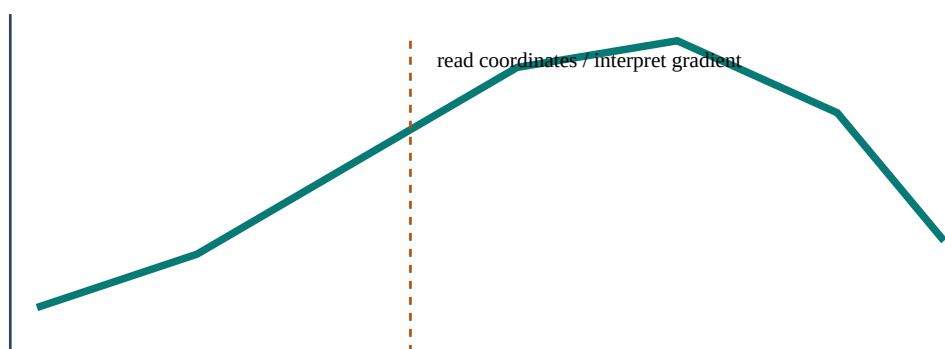
可使用以下骨架组织答案：From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

English explanation. Substitution reverses the chain rule, integration by parts reverses the product rule, and partial fractions turn one rational expression into simpler integrable terms. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Integrate $2x/(x+3)$ 。参考答案：Let $u=x+3, du=dx$; integral $= \ln(x+3) + C$ 。
不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: Let $u=x+3, du=dx$; integral $= \ln(x+3) + C$.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 5 课（逐题精讲） | 复杂积分 | Further integration

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. 计算 $\int x e^x dx$ 。

正确答案 | Correct answer. 取 $u=x, dv=e^x dx$ ，得 $x e^x - \int e^x dx = x e^x - e^x + C$ 。评分展开：

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Integrate $2x/(x+3)$.

参考答案 | Model answer. Let $u=x+3, du=dx$; $\int \frac{2x}{x+3} dx = 2 \int \frac{x+3-3}{x+3} dx = 2 \int \left(1 - \frac{3}{x+3}\right) dx = 2(x+3) - 6 \ln|x+3| + C$ 。得分点：

正确识别对象与条件；选择有效方法；展示关键运算；用完整句回答题目。

第 5 课（精讲续） | 复杂积分 | Further integration

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 6 课 | 微分方程 | Differential equations

Lesson 6 of 10 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

一阶可分离微分方程描述变化率随变量变化；积分常数由初始条件确定。

学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. A differential equation describes a rate law. Separation, integration and an initial condition are three distinct stages, and a lost equilibrium solution must be checked.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

可分离方程把所有 y 项与 dy 放一边，所有 x 项与 dx 放另一边。积分得到的是解族，初始条件确定常数。

公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

把 y 项与 x 项分到两边，分别积分，再代初值。

不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

出现 $1/y$ 积分时严格形式为 $\ln|y|$ ；利用初值和解的符号后，才可写成更简洁指数形式。

若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 6 课（知识点续） | 微分方程 | Differential equations

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

求得的解还应检查定义域、初值以及是否满足原微分方程；平衡解在除以变量时可能被漏掉。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

$dy/dx=3xy$, $y=2$ 当 $x=0$, 求 y 。正确答案: $dy/y=3x dx$; $\ln y=1.5x+C$; $C=\ln 2$, 所以 $y=2e^{(1.5x)}$ 。
完成后逆向代回原条件, 并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

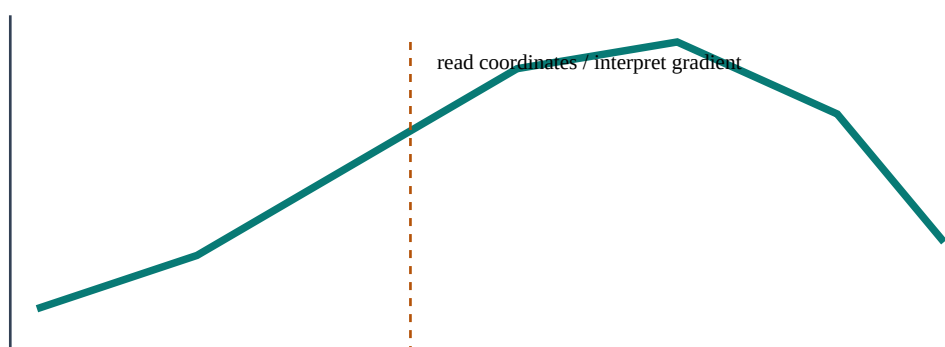
可使用以下骨架组织答案: From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

English explanation. A differential equation describes a rate law. Separation, integration and an initial condition are three distinct stages, and a lost equilibrium solution must be checked. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Why is an arbitrary constant necessary? 参考答案: Differentiation removes constants; integration represents a family of solutions until an initial condition selects one。
不只核对最终数值, 还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: Differentiation removes constants; integration represents a family of solutions until an initial condition selects one.



图表/表示检查 | Representation check

图形必须服务于解题: 标出坐标、参数、区间、交点或关键方向, 并在文字中解释图形与代数/概率表达之间的对应关系。

第 6 课（逐题精讲） | 微分方程 | Differential equations

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. $\frac{dy}{dx}=3xy$, $y=2$ 当 $x=0$, 求 y 。

正确答案 | Correct answer. $\frac{dy}{y}=3x \, dx$; $\ln y=1.5x+C$; $C=\ln 2$, 所以 $y=2e^{(1.5x)}$ 。评分展开:

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Why is an arbitrary constant necessary?

参考答案 | Model answer. Differentiation removes constants; integration represents a family of solutions until an initial condition selects one. 得分点:

正确识别对象与条件; 选择有效方法; 展示关键运算; 用完整句回答题目。

第 6 课（精讲续） | 微分方程 | Differential equations

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 7 课 | 复数 | Complex numbers

Lesson 7 of 10 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

$i=1$; Argand 图把 $z=x+iy$ 视为点; 模与辐角给极形式。

学习时先用自己的话重述定义, 再指出它与相邻概念的区别; 这样遇到改变表述的题目仍能识别本质。

English explanation. Cartesian form displays components; polar form displays magnitude and rotation. Multiplication multiplies moduli and adds arguments, giving complex arithmetic a geometric meaning.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

$z=x+iy$ 在 Argand 图对应点 (x,y) ; 共轭 $z^*=x-iy$; $|z|=\sqrt{x^2+y^2}$, $\arg z$ 是从正实轴量起的角。

公式前先写变量含义和允许范围, 公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

共轭用于实化分母; 乘法在极形式中模相乘、辐角相加。

不要把计算器按键当作证明: 至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

Cartesian form 适合加减; 极形式 $r(\cos \theta + i \sin \theta)$ 适合乘除和幂。除法可乘分母共轭使分母实数化。

若图形趋势、代数符号和数表不一致, 应暂停并检查模型, 而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 7 课（知识点续） | 复数 | Complex numbers

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

辐角应结合象限，不能只写 $\tan(y/x)$ 。根通常等角分布在圆上，必须给出全部不同根。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

化简 $(3+2i)/(1i)$ 。正确答案：乘 $(1+i)$ ： $[(3+2i)(1+i)]/2=(1+5i)/2$ 。
完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

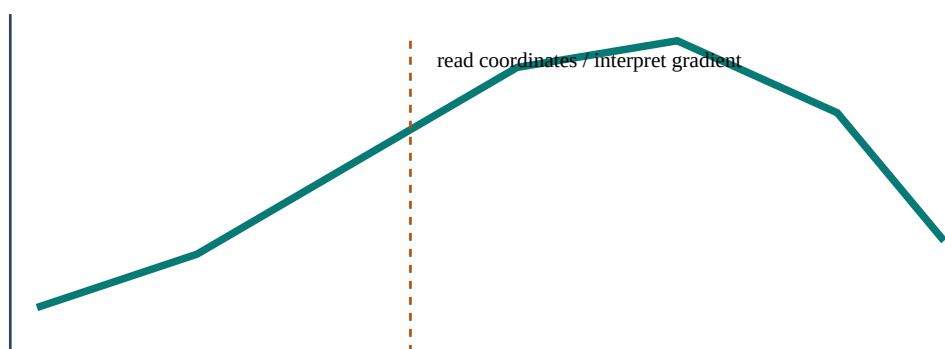
可使用以下骨架组织答案：From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

English explanation. Cartesian form displays components; polar form displays magnitude and rotation. Multiplication multiplies moduli and adds arguments, giving complex arithmetic a geometric meaning. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Find modulus and argument of $1+i$. 参考答案： $|z|=\sqrt{2}$ and $\arg z=\pi/4$ 。
不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: $|z|=\sqrt{2}$ and $\arg z=\pi/4$.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 7 课（逐题精讲） | 复数 | Complex numbers

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. 化简 $(3+2i)/(1i)$ 。

正确答案 | Correct answer. 乘 $(1+i)$: $[(3+2i)(1+i)]/2=(1+5i)/2$ 。评分展开:

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Find modulus and argument of $1+i$.

参考答案 | Model answer. $|z|=\sqrt{2}$ and $\arg z=\pi/4$. 得分点:

正确识别对象与条件; 选择有效方法; 展示关键运算; 用完整句回答题目。

第 7 课（精讲续） | 复数 | Complex numbers

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 8 课 | 向量三维 | Vectors in 3D

Lesson 8 of 10 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

直线向量方程 $r=a+\lambda b$; 交点要求同一点同时满足两条线。

学习时先用自己的话重述定义, 再指出它与相邻概念的区别; 这样遇到改变表述的题目仍能识别本质。

English explanation. A vector equation combines a fixed point with a variable displacement. Intersection requires one common point satisfying every coordinate equation simultaneously.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

位置向量描述点; 方向向量描述直线。 $r=a+\lambda b$ 中 a 是线上一点, b 给方向, λ 取所有实数。

公式前先写变量含义和允许范围, 公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

平行看方向向量成比例; 垂直用点积为0。 不要把计算器按键当作证明: 至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

相交需三个坐标由同一组参数同时满足; 平行看方向向量成比例; 垂直用点积为0。

若图形趋势、代数符号和数表不一致, 应暂停并检查模型, 而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 8 课（知识点续） | 向量三维 | Vectors in 3D

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

两条直线不平行也可能不相交，即 skew lines。只解出两个坐标相同而第三个不符，说明没有交点。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

求(1,2,3)与(4,1,5)距离。正确答案：位移(3,3,2)，距离 $\sqrt{(9+9+4)}=\sqrt{22}$ 。
完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

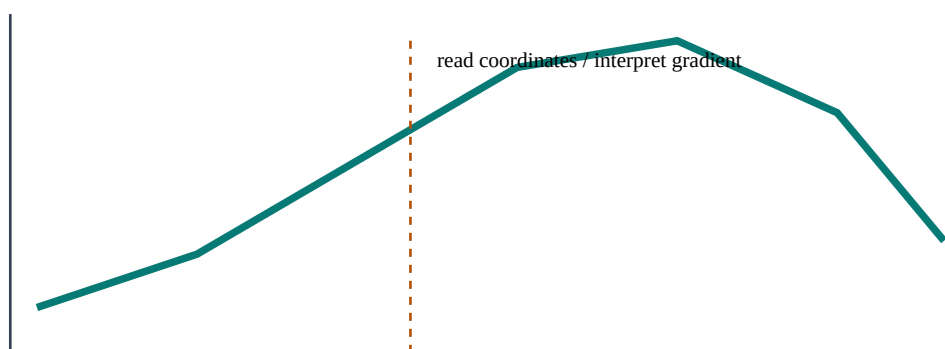
可使用以下骨架组织答案：From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

English explanation. A vector equation combines a fixed point with a variable displacement. Intersection requires one common point satisfying every coordinate equation simultaneously. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Show vectors (1,2,1) and (2,1,0) are perpendicular. 参考答案：Dot product= $2+0=0$ 。
不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: Dot product= $2+0=0$.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 8 课（逐题精讲） | 向量三维 | Vectors in 3D

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. 求 (1,2,3) 与 (4,1,5) 距离。

正确答案 | Correct answer. 位移 (3,3,2)，距离 $\sqrt{(9+9+4)} = \sqrt{22}$ 。评分展开：

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Show vectors (1,2,1) and (2,1,0) are perpendicular.

参考答案 | Model answer. Dot product = $2+0=2 \neq 0$. 得分点：

正确识别对象与条件；选择有效方法；展示关键运算；用完整句回答题目。

第 8 课（精讲续） | 向量三维 | Vectors in 3D

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 9 课 | 数值积分 | Numerical integration

Lesson 9 of 10 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

梯形法用直线段近似曲线下面积，分割越细通常越准确。

学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. The trapezium rule is a weighted sum of ordinates. Its accuracy depends on strip width and curvature; a smaller step usually improves, but does not magically make the result exact.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

梯形法把曲线用连续弦段近似。 $h=(b-a)/n$ ，端点权重1，中间点权重2。

公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

$h=(b-a)/n$ ；端点一次、中间点两次。不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

必须确认 n 是条带数而不是纵坐标数； n 条带需要 $n+1$ 个函数值。列表可减少漏项。

若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 9 课（知识点续） | 数值积分 | Numerical integration

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

曲线凸向上时弦通常在曲线上方，产生高估；凸向下时常为低估。题目若不给形状，不能武断判断误差方向。
Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

用 $n=2$ 梯形估算 $\int x dx$ 。正确答案： $h=1$ ，估算 $=1/2[0+2(1)+4]=3$ 。
完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

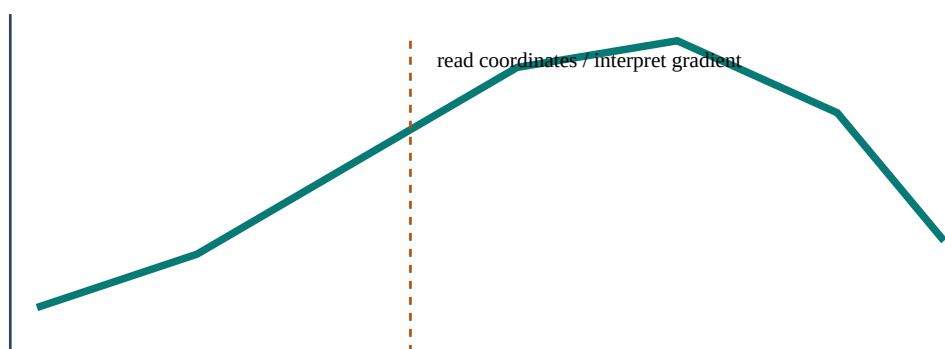
可使用以下骨架组织答案：From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

English explanation. The trapezium rule is a weighted sum of ordinates. Its accuracy depends on strip width and curvature; a smaller step usually improves, but does not magically make, the result exact. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

State why this is an overestimate. 参考答案： For a convex curve $y=x$, the chords lie above the curve.
不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: For a convex curve $y=x$, the chords lie above the curve.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 9 课（逐题精讲） | 数值积分 | Numerical integration

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. 用 $n=2$ 梯形估算 $\int x dx$ 。

正确答案 | Correct answer. $h=1$ ，估算 $=1/2[0+2(1)+4]=3$ 。评分展开：

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. State why this is an overestimate.

参考答案 | Model answer. For a convex curve $y=x$, the chords lie above the curve. 得分点：

正确识别对象与条件；选择有效方法；展示关键运算；用完整句回答题目。

第 9 课（精讲续） | 数值积分 | Numerical integration

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 10 课 | P3 综合策略 | P3 synthesis

Lesson 10 of 10 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

P3 题常让前一小问提供后一问可用的恒等式或分解。

学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. A long Pure Mathematics question is a connected argument. Earlier results are tools, exact forms preserve structure, and domain checks protect the final conclusion.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

P3 综合题常以前一问的恒等式、分解或导数为后一问入口。“Hence”要求使用此前结果，而不是另走冗长路线。公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

先读完整题，保留题设形式；每一步检查定义域、参数与精度。

不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

先通读小问，标记已证明结果的形式；计算过程保留 π 、 e 、 $\ln$ 和根式，最后才按题目要求取近似。

若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 10 课（知识点续） | P3 综合策略 | P3 synthesis

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

任何含参数、分母、对数、根号或反函数的结果都要检查条件。漂亮的代数式若超出定义域仍不是答案。
Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

已知分解后求广义积分时应先检查什么？ 正确答案：
检查分母零点是否落在积分区间；若有，普通定积分可能不存在。
完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

可使用以下骨架组织答案：From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

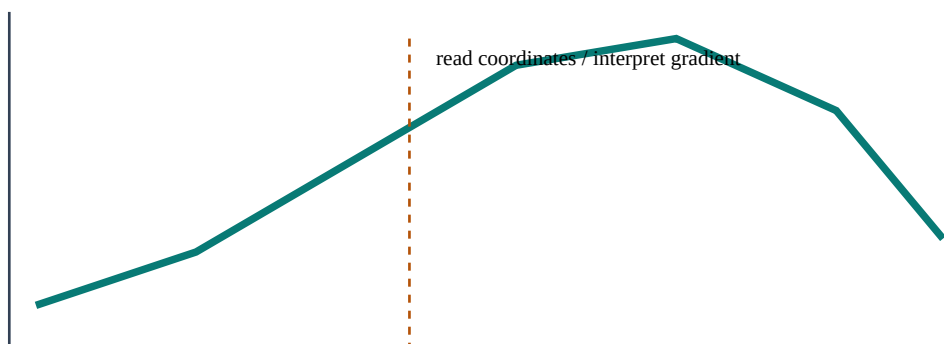
English explanation. A long Pure Mathematics question is a connected argument. Earlier results are tools, exact forms preserve structure, and domain checks protect the final conclusion. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

How should exact answers be presented? 参考答案：Keep π , e , logarithms and radicals exact unless a decimal accuracy is explicitly requested.

不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: Keep π , e , logarithms and radicals exact unless a decimal accuracy is explicitly requested.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 10 课（逐题精讲） | P3 综合策略 | P3 synthesis

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. 已知分解后求广义积分时应先检查什么？

正确答案 | Correct answer. 检查分母零点是否落在积分区间；若有，普通定积分可能不存在。评分展开：

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. How should exact answers be presented?

参考答案 | Model answer. Keep π , e , logarithms and radicals exact unless a decimal accuracy is explicitly requested. 得分点：正确识别对象与条件；选择有效方法；展示关键运算；用完整句回答题目。

第 10 课（精讲续） | P3 综合策略 | P3 synthesis

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.